

Calendar Spreads

Managing The Expiration Cycle

FINM 37000 - Futures And Related Derivatives

Eric Patterson, Fall 2025

Expiration Cycle

Month	Code	Soybeans	Corn	Crude	Ten Years	E-mini S&P	3 Month SOFR
May	K	ZSK5	ZCK5	April Settle			SRK5
June	M			CLM5	ZNM5	ESM5	SR3M5
July	N	ZSN5	ZCN5	CLN5			SRN5
August	Q	ZSQ5		CLQ5			SRQ5
September	U	ZSU5	ZCU5	CLU5	ZNU5	ESU5	SR3U5
October	V			CLV5			SRV5
November	X	ZSX5		CLX5			
December	Z		ZCZ5	CLZ5	ZNZ5	ESZ5	SR3Z5
January	F	ZSF6		CLF6			
February	G			CLG6			
March	H	ZSH6	ZCH6	CLH6	(ZNH6)	ESH6	SR3H6
April	J			CLJ6			
Total		8/15	8/9	10 yrs+2mo		21 quarters	39 qtrs/6 serial

Calendar Spreads

Soybean Example

- ZSN5-ZSQ5
 - Leg 1: +1 ZSN5
 - Leg 2: -1 ZSQ5
- Example: Trade 1 spread for 10'0
 - Typical leg price assignment: most recent price from a leg, say leg 1
 - ZSN5 assigned last traded 1050'0
 - ZSQ5 assigned ZSN5-Spread = 1040'0

Calendar spreads

How trading exchange-listed spreads improves execution

- Suppose hedger wants to sell 50 August soybeans
- Not enough liquidity at the top of the book

• Possibility 1:

- Hit ZSQ5 Bid 1040 for 10
- Hit next 1039'6 for 20
- Hit 1039'4 for 20
- Sold 50 but missed 1040
 - $20 \times 0.25 \times 5000 = \250
 - $20 \times 0.5 \times 5000 = \500

ZSQ5

Bid Quantity	Bid Price	Ask Price	Ask Quantity
		1040'2	10
10	1040'0		
20	1039'6		
20	1039'4		

Calendar spreads

How trading exchange-listed spreads improves execution

- Suppose hedger wants to sell 50 August soybeans, but it's less liquid than July

ZSN5

Bid Quantity	Bid Price	Ask Price	Ask Quantity
100	1050'0		

- Possibility 2:

ZSN5-ZSQ5

Bid Quantity	Bid Price	Ask Price	Ask Quantity
		10'0	50

- Lift 50 spreads offered 10
- Hit 50 July bids 1050'0
- Left short 50 ZSQ5 at 1040

ZSQ5

Bid Quantity	Bid Price	Ask Price	Ask Quantity
10	1040'0		

- Can you execute both trades?

Calendar spreads

How trading exchange-listed spreads improves execution

- Suppose hedger wants to sell 50 August soybeans, but it's less liquid than July

ZSN5

Bid Quantity	Bid Price	Ask Price	Ask Quantity
100	1050'0		

- Possibility 3:

ZSN5-ZSQ5

Bid Quantity	Bid Price	Ask Price	Ask Quantity
		10'0	50

- Exchange implies 50 in ZSQ5
- Hit 1040'0 bid in ZSQ5

ZSQ5

Bid Quantity	Bid Price	Ask Price	Ask Quantity
10 + 50(i)	1040'0		

- First 10 filled by real orders
- Next 40 implied

Calendar spreads

Benefits

- Improve leg execution
 - Often reduced tick size
- Reduced margin requirements (offsetting risk)
- Simplified roll

Calendar Spreads

Rolling Positions

- Suppose you are short 1000 E-mini S&P June futures (ESM5) as a hedge for a portfolio of stocks
- Before those futures expire on June 20th, roll them to September (ESU5)
 - Typical volume in deferred month $< 0.1\%$ of the front month
 - Typical open interest in deferred month $< 2.0\%$ of the front month
 - As June expiration approaches, September volume and spread volume increases

Calendar Spreads

Rolling Positions

- Roll short 1000 S&P E-Mini June futures (ESM5)
- Possibility 1:
 - Lift offer 5500.75 ESM5
 - Closes out ESM5 position
 - Hit bid 5540.75 ESU5

ESM5

Bid Quantity	Bid Price	Ask Price	Ask Quantity
		5500.75	8000

ESU5

Bid Quantity	Bid Price	Ask Price	Ask Quantity
2000	5540.75		

Calendar Spreads

Rolling Positions

- Roll short 1000 S&P E-Mini June futures (ESM5)
- Possibility 2:
 - Hit ESM5-ESU5 bid 40.00
 - Note reversed convention
 - Close ESM5 position
 - Short ESU5 from ESM5+40

ESM5

Bid Quantity	Bid Price	Ask Price	Ask Quantity
		5500.75	8000

ESM5-ESU5

Bid Quantity	Bid Price	Ask Price	Ask Quantity
4000	40.00		

ESU5

Bid Quantity	Bid Price	Ask Price	Ask Quantity
2000	5540.75		

Calendar Spreads

Rolling Positions

- Roll short 1000 S&P E-Mini June futures (ESM5)

ESM5

Bid Quantity	Bid Price	Ask Price	Ask Quantity
		5500.75	8000

- Possibility 3: Reduced tick size

- Hit ESM5-ESU5 bid 40.15

ESM5-ESU5

Bid Quantity	Bid Price	Ask Price	Ask Quantity
4000	40.15	40.20	4000

- Short ESU5 from ESM5+40.15

- Improved execution:

ESU5

Bid Quantity	Bid Price	Ask Price	Ask Quantity
2000	5540.75		

- $1000 \times 0.15 \times \$50 = \$7500$

Other baskets

Reducing execution costs for multi-leg positions

- Packs
 - 4 consecutive quarterly futures
 - Price is relative to each futures' previous settle
- Bundles
 - At least 4 consecutive quarterly futures
 - Price as average of the futures
- Strips
 - At least 2 legs of same product with different expirations
 - Various pricing styles